EE 121B: Principles of Semiconductor Device Design

Lecture 7:

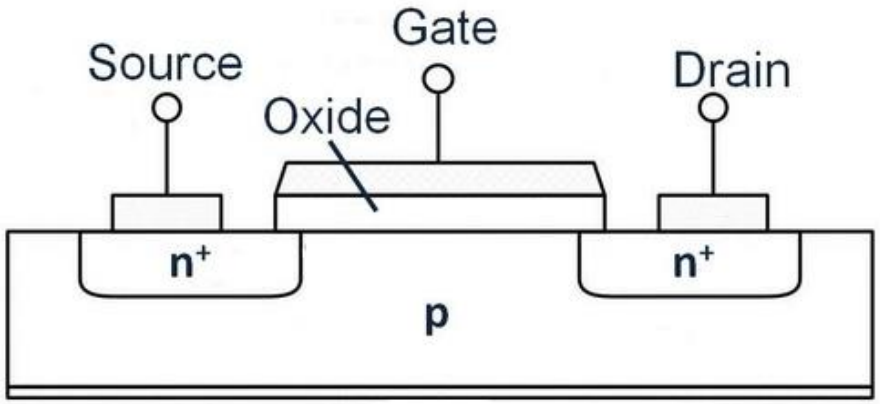
pn Junction Diodes (continued)

Sam Emaminejad

Course Overview

1. Semiconductor Properties
2. Metal-Semiconductor Contacts
3. P-N Junction
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



http://qph.is.quoracdn.net/main-qimg-38a99b4f215e161bf31964795cfce7d2?convert_to_webp=true

Lecture 7

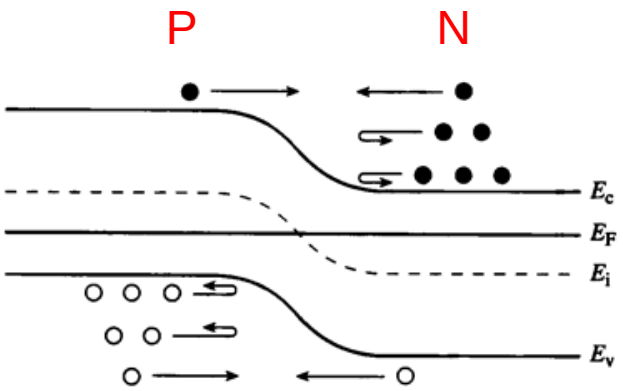
OUTLINE

- pn Junction Diodes (cont'd)
  - Derivation of the Ideal Diode Equation (for a step junction)

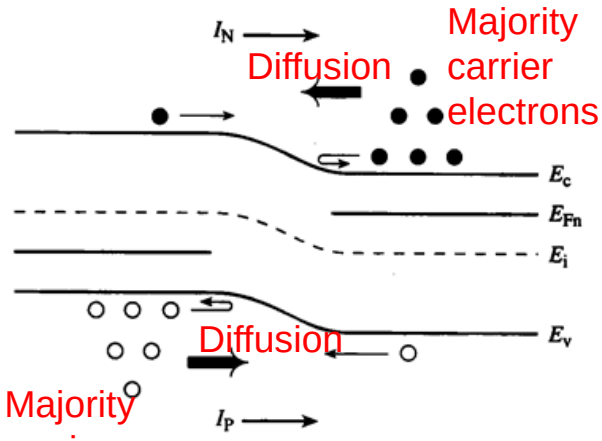
Reading: Pierret 6.1; Hu 4.3, 4.6, 4.8-4.9

Current Flow (Qualitative View)

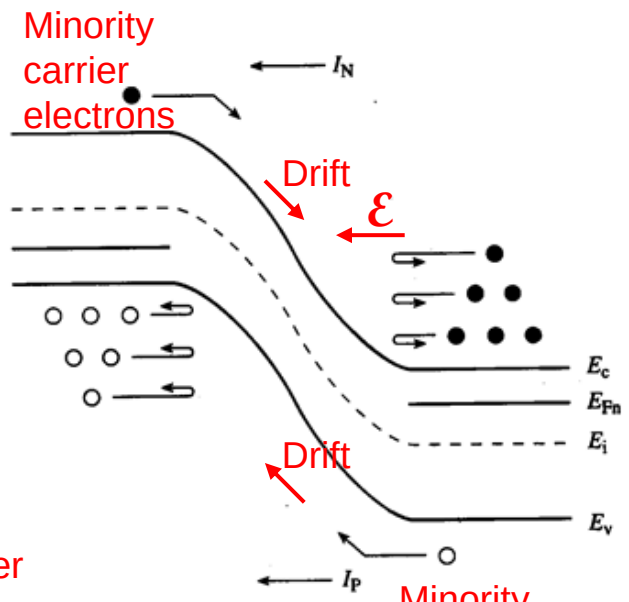
Equilibrium ($V_A = 0$)

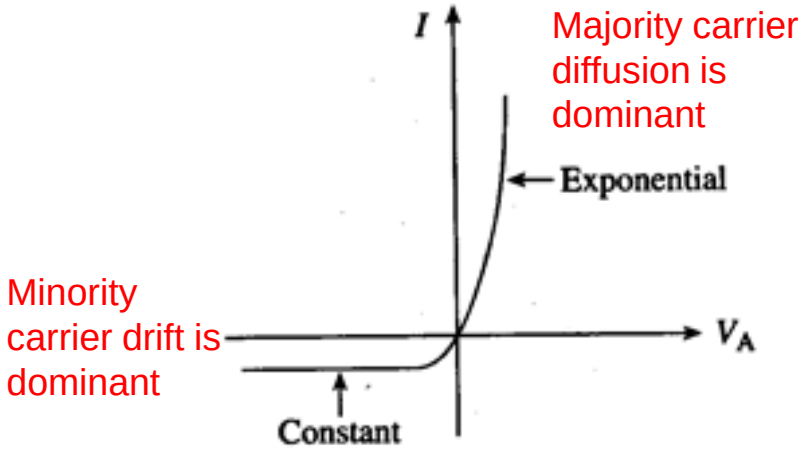


Forward Bias ($V_A > 0$)



Reverse Bias ($V_A < 0$)

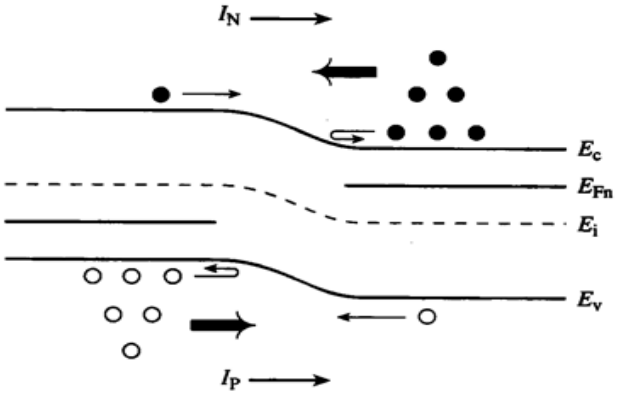




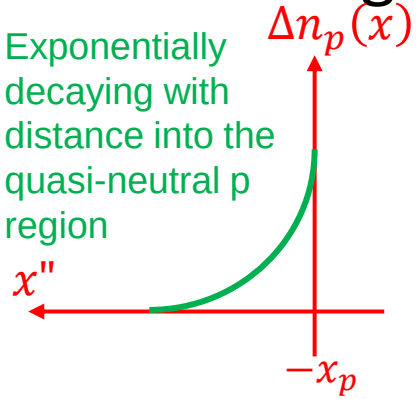
R. F. Pierret, *Semiconductor Device Fundamentals*, pp. 236-237

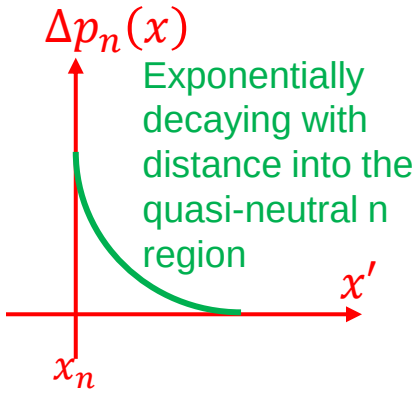
Carrier Action under Forward Bias

- When a forward bias (V_A>0) is applied, the potential barrier to diffusion across the junction is reduced
  - Minority carriers are "injected" into the quasi-neutral regions => $\Delta n_p > 0$, $\Delta p_n > 0$



- Minority carriers diffuse in the quasi-neutral regions, recombining with majority carriers





Ideal Diode Analysis: Assumptions

- Non-degenerately doped step junction
- Steady-state conditions $\frac{dn}{dt} = 0$ $\frac{d\rho}{dt} = 0$
- Low-level injection conditions in quasi-neutral regions

$$n_n = n_{n0} = N_D \qquad p_p = p_{p0} = N_A$$

- Recombination-generation negligible in depletion region

$$\Rightarrow \quad \frac{dJ_n}{dx} = 0, \quad \frac{dJ_p}{dx} = 0$$

i.e. J_n & J_p are constant inside the depletion region

Components of Current Flow

- **Current density $J = J_n(x) + J_p(x)$**

drift diffusion

$n = n_0 + \Delta n \rightarrow \frac{dn}{dx} = \frac{d\Delta n}{dx}$

Electron: $$J_n(x) = q\mu_n n\mathcal{E} + qD_n \frac{dn}{dx} = q\mu_n n\mathcal{E} + qD_n \frac{d(\Delta n)}{dx}$$

Hole: $$J_p(x) = q\mu_p p\mathcal{E} - qD_p \frac{dp}{dx} = q\mu_p p\mathcal{E} - qD_p \frac{d(\Delta p)}{dx}$$

$p = p_0 + \Delta p \rightarrow \frac{dp}{dx} = \frac{d\Delta p}{dx}$

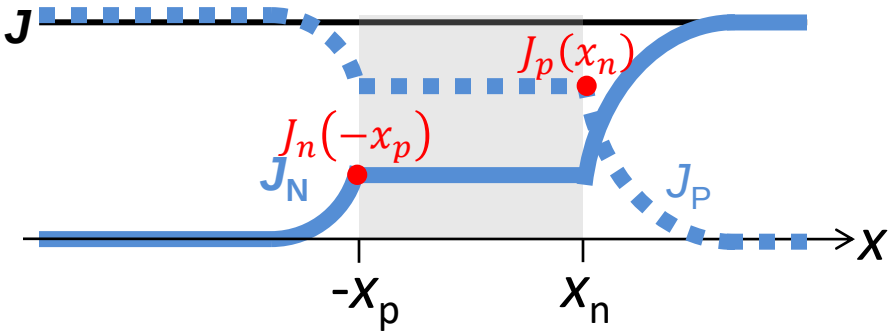
- ***J* is constant throughout the diode, but $J_n(x)$ and $J_p(x)$ vary with position:**

$J_n + J_p = J$ throughout (for all x)

Example:

p⁺n junction under forward bias:

$J_p > J_n$ across the junction



"Game Plan" for Obtaining Diode *I-V*

1. Solve minority-carrier diffusion equations in quasi-neutral regions to obtain excess carrier distributions $\Delta n_p(x,V_A)$,$\Delta p_n(x,V_A)$
   - boundary conditions:
     - p side: $\Delta n_p(-x_p)$, $\Delta n_p(-\infty)$ $= 0 \rightarrow \Delta n_p(x)$
     - n side: $\Delta p_n(x_n)$, $\Delta p_n(\infty)$ $= 0 \rightarrow \Delta p_n(x)$
2. Find minority-carrier current densities in quasi-neutral regions

$$J_n(x,V_A) = qD_n \frac{d(\Delta n_p)}{dx} \qquad J_p(x,V_A) = -qD_p \frac{d(\Delta p_n)}{dx}$$

3. Evaluate J_n at $x=-x_p$ & J_p at $x=x_n$ to obtain total current density J:

$$J(V_A) = J_n(-x_p,V_A) + J_p(x_n,V_A)$$

Carrier Concentrations at $-x_p$, x_n

Consider the **equilibrium** ($V_A = 0$) carrier concentrations:

p side

$$p_{p0}(-x_p) = N_A$$

$$n_{p0}(-x_p) = \frac{n_i^2}{N_A}$$

n side

$$n_{n0}(x_n) = N_D$$

$$p_{n0}(x_n) = \frac{n_i^2}{N_D}$$

If low-level injection conditions hold in the quasi-neutral regions when $V_A \neq 0$, then

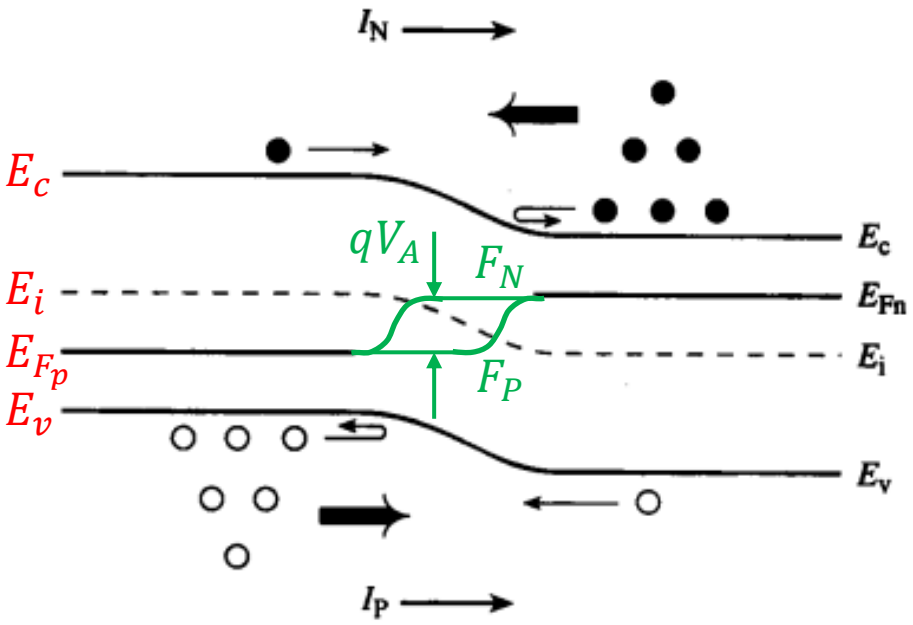
$$p_p(-x_p) = N_A$$

$$n_n(x_n) = N_D$$

“Law of the Junction”

The voltage applied to a pn junction falls mostly across the depletion region (assuming low-level injection in the quasi-neutral regions).

We can draw 2 quasi-Fermi levels in the depletion region:



$$p = n_i e^{(E_i - F_P)/kT}$$

$$n = n_i e^{(F_N - E_i)/kT}$$

$$pn = n_i^2 e^{(F_N - F_P)/kT}$$

$$\boxed{pn = n_i^2 e^{qV_A/kT}}$$

In the depletion region

Excess Carrier Concentrations at $-x_p$, x_n

p side

$$p_p(-x_p) = N_A$$

$$n_p(-x_p) = \frac{n_i^2 e^{qV_A/kT}}{N_A}$$

$$= n_{p0} e^{qV_A/kT}$$

$$\Delta n_p = n_p - n_{p0} = n_{p0} e^{\frac{qV_A}{kT}} - n_{p0}$$

$$\Delta n_p(-x_p) = \frac{n_i^2}{N_A}\left(e^{qV_A/kT} - 1\right)$$

n side

$$n_n(x_n) = N_D$$

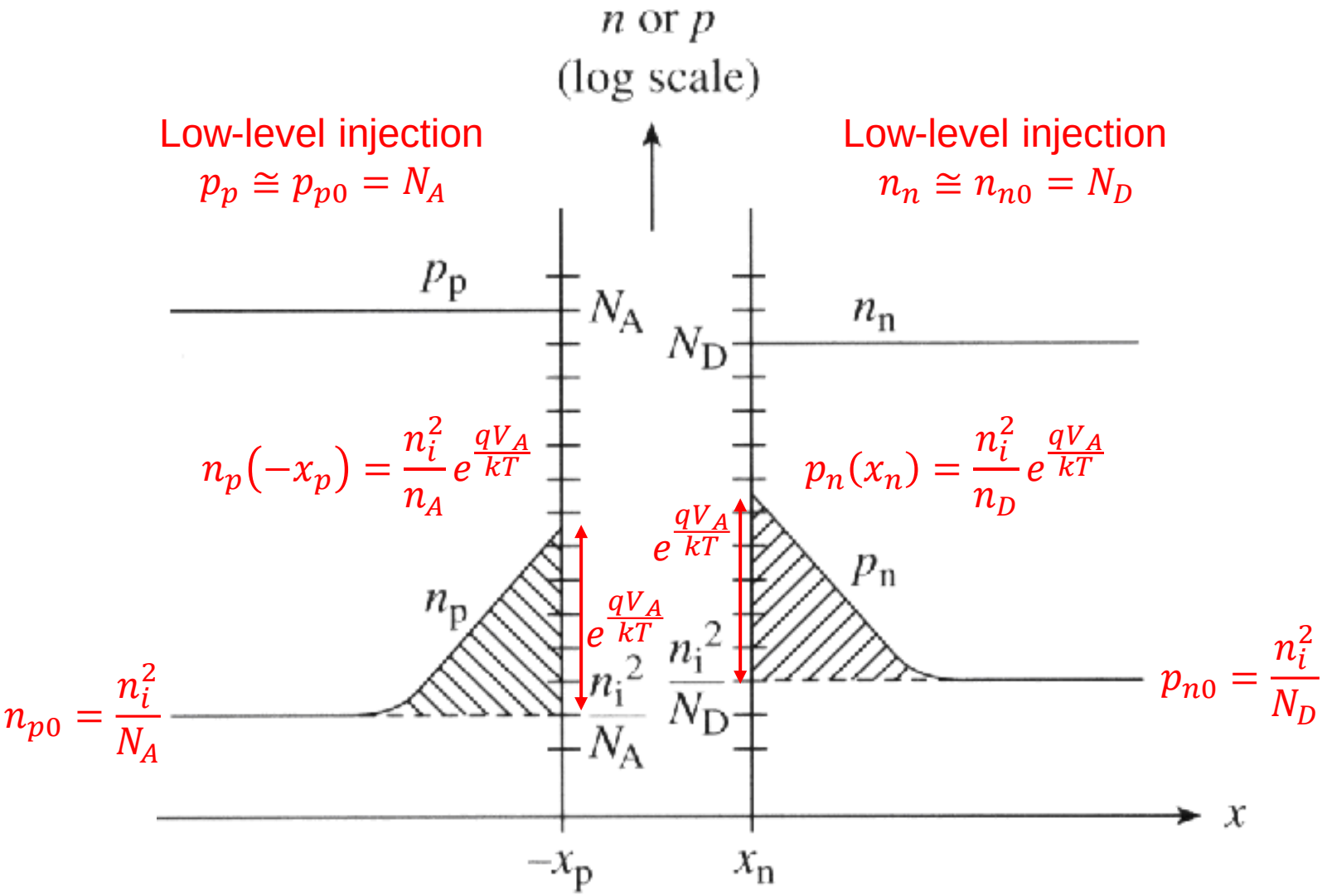
$$p_n(x_n) = \frac{n_i^2 e^{qV_A/kT}}{N_D}$$

$$= p_{n0} e^{qV_A/kT}$$

$$\Delta p_n = p_n - p_{n0} = p_{n0} e^{\frac{qV_A}{kT}} - p_{n0}$$

$$\Delta p_n(x_n) = \frac{n_i^2}{N_D}\left(e^{qV_A/kT} - 1\right)$$

Carrier Concentration Profiles under Forward Bias



R. F. Pierret, *Semiconductor Device Fundamentals*, Fig. 6.8a

Example

Since $\frac{kT}{q}\ln 10 = 0.06\text{V}$

$0.6\text{V} = 10 \times \frac{kT}{q}\ln 10 = \frac{kT}{q}\ln 10^{10}$

Consider a pn junction with N_A=10^{18} cm^{-3} and N_D=10^{16} cm^{-3}, under a forward bias of 0.6 V.

$e^{\frac{qV_A}{kT}} = e^{\frac{0.6}{kT/q}} = 10^{10}$

(a) What are the minority carrier concentrations at the edges of the depletion region?

$$n_p(-x_p) = \frac{n_i^2}{N_A} e^{\frac{qV_A}{kT}} = \underset{n_{p0}}{10^2}(10^{10}) = 10^{12}\text{cm}^{-3}$$

$$p_n(+x_n) = \frac{n_i^2}{N_D} e^{\frac{qV_A}{kT}} = 10^4(10^{10}) = \underset{p_{n0}}{10^{14}}\text{cm}^{-3}$$

(b) What are the excess minority carrier concentrations at the edges of the depletion region?

$$\Delta n_p(-x_p) = n_p(-x_p) - n_{p0} = 10^{12} - 10^2 = 10^{12}\text{cm}^{-3}$$

$$\Delta p_n(x_n) = p_n(x_n) - p_{n0} = 10^{14} - 10^4 = 10^{14}\text{cm}^{-3}$$

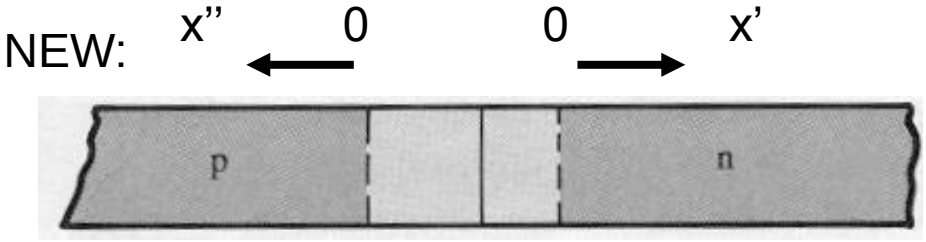
Excess Carrier Distribution (n side)

- From the minority carrier diffusion equation: $\frac{d^2\Delta p_n}{dx^2} = \frac{\Delta p_n}{D_p \tau_p} = \frac{\Delta p_n}{L_p^{\ 2}}$

- We have the following boundary conditions:

$$\Delta p_n(x_n) = p_{no}(e^{qV_A/kT} - 1) \qquad \Delta p_n(\infty) \to 0$$

- For simplicity, use a new coordinate system:



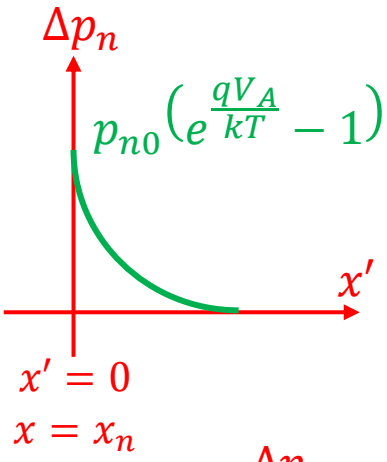
- Then, the solution is of the form: $\Delta p_n(x') = A_1 e^{x'/L_p} + A_2 e^{-x'/L_p}$

$$\Delta p_n(x') = A_1 e^{x'/L_p} + A_2 e^{-x'/L_p}$$

From the $x = \infty$ boundary condition:

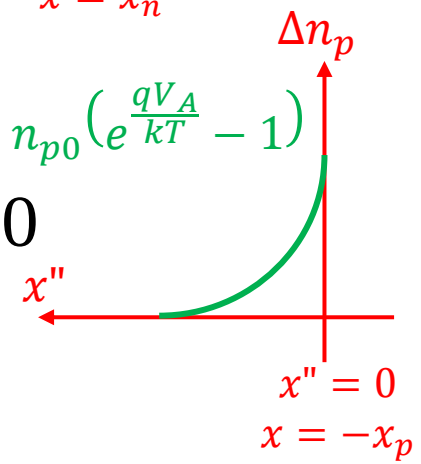
From the $x = x_n$ boundary condition:

Therefore $\Delta p_n(x') = p_{no}(e^{qV_A/kT} - 1)e^{-x'/L_p}, \; x' > 0$



Similarly, we can derive

$$\Delta n_p(x'') = n_{po}(e^{qV_A/kT} - 1)e^{-x''/L_n}, \; x'' > 0$$



Total Current Density

p side: $$J_n = -qD_n \frac{d\Delta n_p(x'')}{dx''} = q\frac{D_n}{L_n} n_{p0}(e^{qV_A/kT} - 1)e^{-x''/L_n}$$

($n_{p0} = n_i^2/N_A$)

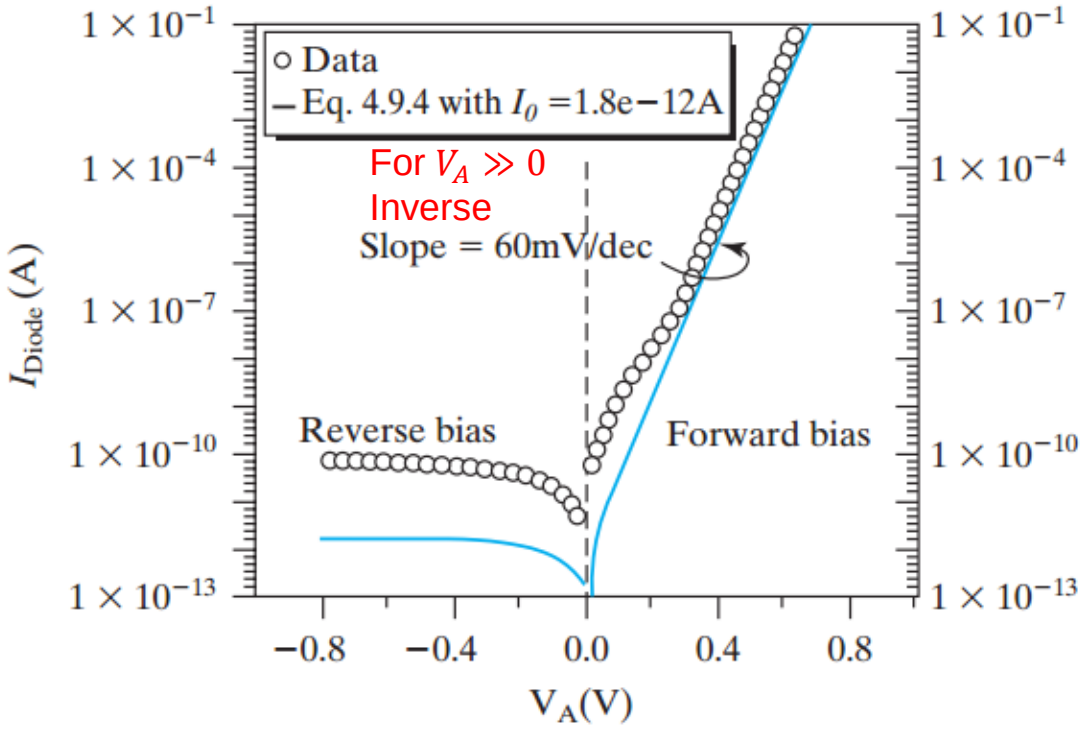
n side: $$J_p = -qD_p \frac{d\Delta p_n(x')}{dx'} = q\frac{D_p}{L_p} p_{n0}(e^{qV_A/kT} - 1)e^{-x'/L_p}$$

($p_{n0} = n_i^2/N_D$)

$$J = J_n\Big|_{x=-x_p} + J_p\Big|_{x=x_n} = J_n\Big|_{x''=0} + J_p\Big|_{x'=0}$$

$$J = qn_i^2\left[\frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D}\right](e^{qV_A/kT} - 1) \quad [\text{Amperes/cm}^2]$$

Ideal Diode Equation



C. C. Hu, *Modern Semiconductor Devices for Integrated Circuits*, Figure 4-22

$$I = I_0(e^{qV_A/kT} - 1) \text{ [Amperes]}$$

$$I_0 = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)$$

Hole diffusion across junction

Electron diffusion across junction

Suppose $I_1 = I_0\left(e^{\frac{qV_{A1}}{kT}} - 1\right) \cong I_0 e^{\frac{qV_{A1}}{kT}}$ and $I_2 = I_0\left(e^{\frac{qV_{A2}}{kT}} - 1\right) \cong I_0 e^{\frac{qV_{A2}}{kT}}$

If $\frac{I_2}{I_1} = 10$, then $\frac{e^{qV_{A2}/kT}}{e^{qV_{A1}/kT}} = 10 \rightarrow e^{q(V_{A2}-V_{A1})/kT} = 10 \rightarrow V_{A2} - V_{A1} = \frac{kT}{q}\ln 10 = 0.06\text{V}$

1 decade change in current ↔ 60 mV change in V_A

Diode Saturation Current I_0

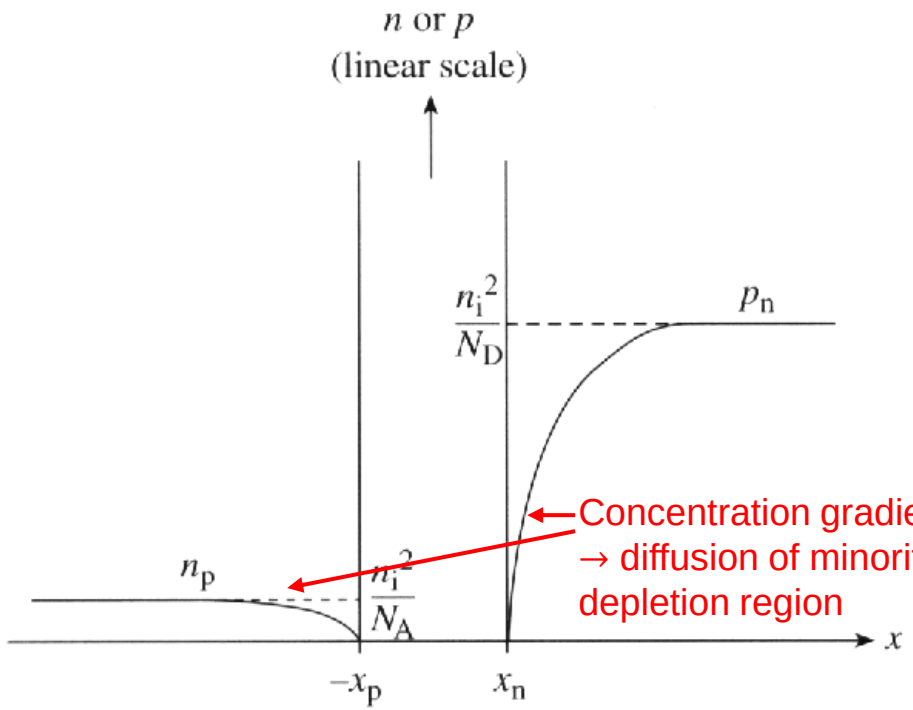
- I_0 can vary by orders of magnitude, depending on the semiconductor material and dopant concentrations:

$$I_0 = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)$$

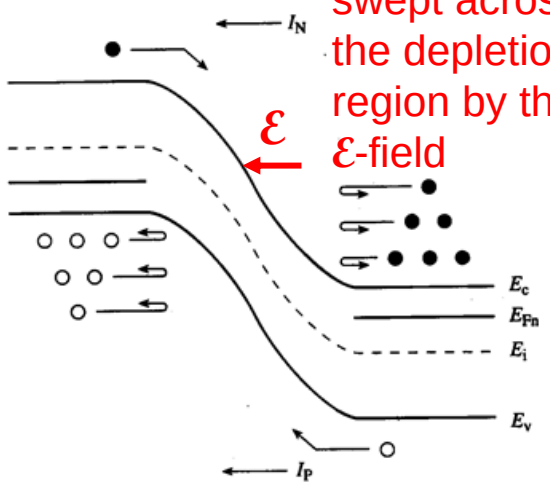
- In an asymmetrically doped (one-sided) pn junction, the term associated with the more heavily doped side is negligible:
  - If the p side is much more heavily doped, $I_0 \cong Aqn_i^2\left(\frac{D_p}{L_pN_D}\right)$
  - If the n side is much more heavily doped, $I_0 \cong Aqn_i^2\left(\frac{D_n}{L_nN_A}\right)$

Carrier Concentration Profiles under Reverse Bias

R. F. Pierret, *Semiconductor Device Fundamentals*, Fig. 6.8b



Concentration gradient → diffusion of minority carriers to depletion region



Minority carriers are swept across the depletion region by the $\mathcal{E}$-field

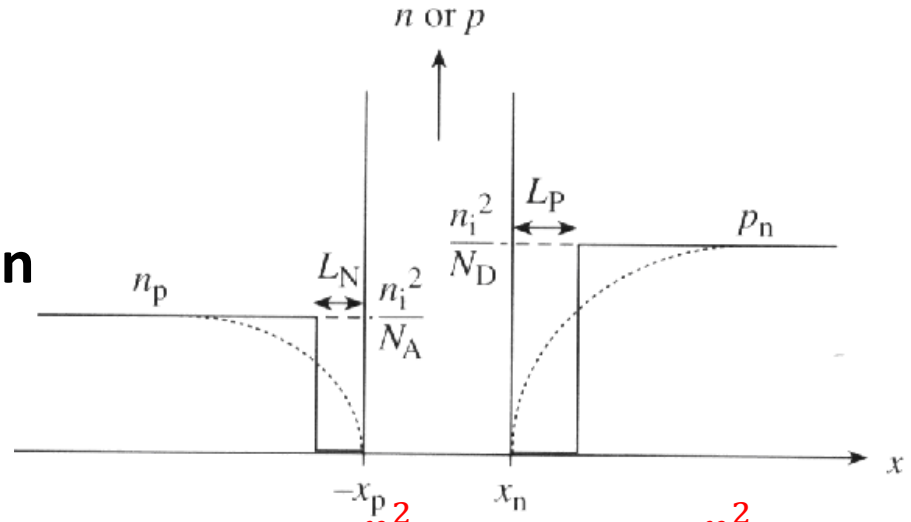
Depletion of minority carriers within ~1 diffusion length of depletion region

- Depletion of minority carriers at edges of depletion region
- The only current which flows is due to **drift** of minority carriers across the junction. This current is fed by diffusion of minority carriers toward junction (supplied by thermal generation).

Alternative Derivation of Formula for I_0

"Depletion approximation":

- **I_0 is the rate at which carriers are thermally generated within one diffusion length of the depletion region:**



$$\frac{\partial n}{\partial t} = -\frac{\Delta n_p}{\tau_n} = \frac{n_i^{\,2}/N_A}{\tau_n} \qquad -L_N \le x \le -x_p$$

$$\frac{\partial p}{\partial t} = -\frac{\Delta p_n}{\tau_p} = \frac{n_i^{\,2}/N_D}{\tau_p} \qquad x_n \le x \le x_n + L_P$$

$\Delta n_p = -\frac{n_i^2}{N_A}$ in this region $\qquad \Delta p_n = -\frac{n_i^2}{N_D}$ in this region

Note that $\frac{L_n}{\tau_n} = \frac{D_n}{L_n}$ and $\frac{L_p}{\tau_p} = \frac{D_p}{L_p}$

$$I_0 = qAL_N\left(\frac{n_i^{\,2}/N_A}{\tau_n}\right) + qAL_P\left(\frac{n_i^{\,2}/N_D}{\tau_p}\right) = qAn_i^2\left(\frac{D_n}{L_N N_A} + \frac{D_p}{L_p N_D}\right)$$

R. F. Pierret, *Semiconductor Device Fundamentals*, Fig. E6.4

Summary

- Under forward bias ($V_A > 0$), the potential barrier to carrier diffusion is reduced → minority carriers are "injected" into the quasi-neutral regions.
  - The minority-carrier concentrations at the edges of the depletion region change with the applied bias V_A, by the factor $e^{qV_A/kT}$
  - The excess carrier concentrations in the quasi-neutral regions decay to zero away from the depletion region, due to recombination.

pn junction diode current $I = qAn_i^2\left[\frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D}\right](e^{qV_A/kT} - 1)$

- I_0 can be viewed as the drift current due to minority carriers generated within a diffusion length of the depletion region

Acknowledgement

- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
- Professor Tsu-Jae King Liu EE130/230A notes